

Alex Sjothun

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Summary

Senior software engineer with 8+ years of experience designing Python-based, data-driven applications, analytical platforms, and decision-support tools. Expert in Python, SQL, relational databases, and building robust data pipelines and processing workflows using modern frameworks such as Flask, FastAPI, Dash, and Streamlit. Proven track record delivering secure, scalable analytics services, interactive dashboards, and simulation-style scenario analysis in fast-paced, stakeholder-facing environments. Strong focus on application architecture, CI/CD, automated testing, and clear documentation to translate complex analytical requirements into maintainable production solutions.

Skills

- **Programming Languages:** Python, SQL, JavaScript, TypeScript, HTML5, CSS3, Bash, Java
- **Python Frameworks Libraries:** Flask, FastAPI, Dash, Streamlit, Pandas, NumPy, SQLAlchemy, Pydantic
- **Data Analytics:** Data pipelines, ETL development, Data modeling, Relational databases, Data validation, Analytical application design, Scenario-based analysis, Simulation-style modeling
- **Databases Storage:** PostgreSQL, MySQL, SQLite, Cloud SQL, Database schema design, Query optimization, Indexing strategies, Data access layer design
- **Web Frontend:** RESTful API design, Responsive UI development, Plotly, Data visualization, Interactive dashboards, React, Component-based UI, User-centric design
- **DevOps Quality:** Git, CI/CD pipelines, GitHub Actions, Automated testing, Pytest, Unit and integration testing, Code reviews, Application performance optimization
- **Architecture Practices:** Application architecture, Microservices-style services, Secure coding practices, Dependency management, Observability and logging, Requirements gathering, Technical documentation, Agile collaboration

Experience

Sep 2022 – Feb 2026 (3 years 5 months)

Breezy - Senior Software Engineer, Backend

- Led design and implementation of Python-based analytics services using FastAPI and Flask, exposing RESTful endpoints that powered internal decision-support dashboards and tools.
- Built robust ETL-style data pipelines in Python with Pandas and SQLAlchemy, handling ingestion, transformation, validation, and loading into PostgreSQL for downstream analytics workloads.
- Designed and optimized complex SQL queries and relational schemas in PostgreSQL, improving key analytical report performance by over 40 percent through indexing and query refactoring.
- Developed interactive dashboards with Dash and Plotly to visualize operational metrics, enabling product and operations teams to explore trends and drill into granular data views.
- Implemented a modular application architecture separating API, data access, and analytics layers, improving maintainability and enabling safer, incremental feature delivery.
- Introduced automated testing with Pytest and factory-based fixtures, raising unit and integration test coverage for core analytics services to above eighty percent across the codebase.
- Set up CI/CD pipelines with GitHub Actions to run tests, linting, and security checks on every change, ensuring consistent deployment quality for analytics-focused microservices.
- Collaborated closely with data analysts and business stakeholders to translate reporting and planning requirements into scalable Python services and SQL-based data models.
- Implemented secure coding practices including parameterized queries, strict input validation, and dependency scanning, reducing vulnerabilities in data-driven endpoints and workflows.
- Optimized data processing workflows by batching operations and tuning connection pooling, cutting nightly

analytics job runtimes from hours to under forty minutes in production.

- Built Streamlit-based internal tools for exploratory analysis and ad hoc reporting, allowing non-technical users to self-serve insights without direct database access or manual exports.
- Documented application architecture, data flow diagrams, and operational runbooks, improving onboarding speed for new engineers and clarifying ownership of analytics components.
- Partnered with infrastructure engineers to containerize analytics applications using Docker and refine deployment configurations for reliability, observability, and rollback safety.
- Implemented structured logging and metrics collection for analytics APIs, enabling faster troubleshooting of data quality issues and performance bottlenecks in production environments.
- Mentored mid-level engineers on Python best practices, SQL optimization, and analytical application design, raising overall team competency in data-centric software engineering.

Oct 2020 – Aug 2022 (1 year and 10 months)

Thread - *Senior Software Engineer, Data*

- Designed and delivered Python-based data services that powered reporting and planning features, using Flask and FastAPI to expose secure APIs for internal analytical consumers.
- Developed end-to-end data pipelines for ingestion, transformation, and validation of customer and behavioral data, leveraging Pandas and SQLAlchemy with PostgreSQL as the core store.
- Created reusable SQL views and stored logic to support analytics queries, improving consistency of business metrics and reducing duplicated query logic across reporting tools.
- Built interactive analytics dashboards with Dash and Plotly to visualize user engagement and operational KPIs, enabling product managers to run scenario-style analyses on demand.
- Collaborated with analytics and product teams to refine complex requirements into maintainable schemas, Python modules, and API contracts aligned with long-term application architecture.
- Implemented automated regression and integration tests around critical data workflows using Pytest, catching data quality regressions early in the development lifecycle.
- Introduced CI workflows with Git-based branching strategies and pull request checks, standardizing code reviews and improving reliability of deployments for analytics features.
- Optimized SQL queries and indexing strategies for high-volume tables, reducing latency for key dashboard queries and improving responsiveness of analytical interfaces for users.
- Developed internal Streamlit apps for exploratory analysis and quick what-if evaluations, giving stakeholders a lightweight interface for data exploration without engineering support.
- Ensured secure handling of sensitive data by enforcing least-privilege database roles, encrypted connections, and careful masking of fields in logs and diagnostic outputs.
- Documented data models, pipeline dependencies, and operational procedures in a shared knowledge base, improving transparency and reducing onboarding time for new team members.
- Worked in an agile, cross-functional environment, frequently partnering with client-facing teams to troubleshoot analytics issues and prioritize enhancements to decision-support tools.

Oct 2017 – Sep 2020 (3 years)

Newsela – *Software Engineer*

- Implemented backend features in Python for content and usage analytics, contributing to data-driven insights that supported instructional decision-making in the education domain.
- Developed SQL queries and reporting views on top of MySQL and PostgreSQL, enabling learning teams to track engagement metrics and refine content strategies based on data.
- Assisted in building early-stage data pipelines for aggregating event data, using Python scripts and scheduled jobs to prepare datasets for downstream analytical consumption.
- Contributed to internal dashboards that surfaced key usage statistics, collaborating with product managers to ensure

visualizations aligned with educator workflows and needs.

- Worked on RESTful APIs using Flask to expose analytics-related endpoints, focusing on clear contracts, pagination, and efficient data access patterns for client applications.
- Participated in code reviews and refactoring efforts to improve Python code quality, modularity, and adherence to established application architecture guidelines across services.
- Wrote unit and integration tests with Pytest to validate business logic and data access layers, helping maintain reliability as new analytics features were introduced.
- Collaborated with data and analytics teams to clarify requirements, refine SQL logic, and validate that reported metrics matched expectations from stakeholders and educators.
- Helped troubleshoot performance issues in reporting queries by analyzing execution plans and suggesting indexing or query rewrites to reduce response times for dashboards.
- Documented service endpoints, data contracts, and operational runbooks, supporting smoother handoffs between engineering, analytics, and support teams for data-centric features.

Education

2013 - 2017

University of Texas at Austin, Austin, TX

Bachelor's degree, Computer Science

Certifications

AWS SimuLearn: Generative AI Architect

AWS SimuLearn: Machine Learning

AWS SimuLearn: Solutions Architect